



3	0.4 lag		
4	0.6 lag		
5	0.8 lag		
6	1		
7	0.8 lead		
8	0.6 lead		
9	0.4 lead		
10	0.2 lead		
11	0 lead		

$$\bar{V} = V \angle 0^\circ = \underline{\hspace{2cm}} V$$

$$\bar{I} = I \angle (-\phi)^\circ = \underline{\hspace{2cm}} A$$

$$\bar{E}_f = V \angle 0^\circ + I \angle (-\phi)^\circ \times (R_a + jX_s) = E_f \angle (-\delta)^\circ = \underline{\hspace{2cm}} V$$

$$\text{Or } E_f = \sqrt{(V \cos \phi + I_a R_a)^2 + (V \sin \phi + I_a X_s)^2} = \underline{\hspace{2cm}} V$$

$$\% \text{ regulation} = \frac{E_f - V}{V} \times 100 \% = \underline{\hspace{2cm}} \%$$

PHASOR DIAGRAM (EMF METHOD)

